Hao Chen

CONTACT INFORMATION	West Lafayette, IN, USA		chen4433@purdue.edu
	Google Scholar	Homepage	Linkedin

EDUCATION

Purdue University, West Lafayette, IN
Ph.D. in Chemical Engineering — 2022 - Present

- Research advisor: Dr. Can Li
- Research expertise: Machine Learning and Mathematical Optimization

University of Cambridge, Cambridge, UK
M.Phil. in Chemical Engineering and Biotechnology — 2020 - 2021

University of Nottingham, Nottingham, UK
B.Eng. in Chemical Engineering — 2016 - 2020

APPOINTMENTS

Amazon, Bellevue, WA — 2025
Applied Scientist Intern, Last Mile Planning

PUBLICATIONS

[1] GE Constante-Flores, **H Chen**, C Li. Enforcing Hard Linear Constraints in Deep Learning Models with Decision Rules. *arXiv.* (2025).

[2] A Ramanujam, A Elyoumi, H Chen, SM Kompalli, AS Ahluwalia, S Pal, DJ Papageorgiou, C Li. SafeOR-Gym: A Benchmark Suite for Safe Reinforcement Learning Algorithms on Practical Operations Research Problems. *arXiv.* (2025).

[3] **H Chen**, GE Constante-Flores, KSI Mantri, SM Kompalli, AS Ahluwalia, C Li. OptiChat: Bridging Optimization Models and Practitioners with Large Language Models. *arXiv.* (2025).

[4] A Khan, R Nahar, H Chen, GE Constante-Flores, C Li. FaultExplainer: Leveraging Large Language Models for Interpretable Fault Detection and Diagnosis. *Computers & Chemical Engineering.* (2024).

[5] YC Lu, A Varghese, R Nahar, H Chen, K Shao, X Bao, C Li. scChat: A Large Language Model-Powered Co-Pilot for Contextualized Single-Cell RNA Sequencing Analysis. *bioRxiv.* (2024).

[6] **H Chen**, GE Constante-Flores, C Li. Physics-informed neural networks with hard linear equality constraints. *Computers & Chemical Engineering 189, 108764.* (2024).

[7] **H Chen**, GE Constante-Flores, C Li. Diagnosing Infeasible Optimization Problems Using Large Language Models. *INFOR: Information Systems and Operational Research, 1–15.* (2024).

[8] Y Yan, WI Shin, H Chen, et al. A recent trend: application of graphene in catalysis. *Carbon Lett. 31, 177–199* (2021).

CONFERENCE PRESENTATIONS

Chen, H. 2024. Deep Learning Models with Hard Linear Inequality Constraints. Paper presented at the *2025 AIChE Annual Meeting*, Boston, MA

Chen, H. 2024. Self-supervised Learning for Constrained Optimization with Hard Linear Constraints. Paper presented at the *2024 INFORMS Annual Meeting*, Seattle, WA

Chen, H. 2024. GPU Accelerated Approximation Algorithm for Multi-Parametric Linear Programming. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Chen, H. 2024. Physics-Informed Neural Networks with Hard Linear Equality Constraints. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Chen, H. 2024. Diagnosing Infeasible Optimization Problems Using Large Language Models. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Honors and Awards

Ryland Travel Grant, School of Chemical Engineering, Purdue University	2025
Purdue Graduate Student Government (PGSG) Travel Grant	2024
Chao Travel Grant, School of Chemical Engineering, Purdue University	2024
Nottingham Engineering Excellence Scheme (Top 1.0%), University of Nottingham	2019
British Petroleum Prize, University of Nottingham	2019
Provost's Scholarship (Top 1.5%), University of Nottingham	2018
Dean's Scholarship (Top 10%), University of Nottingham	2017

Research Mentoring

Priya Kumari (M.S. from Purdue)
Ashley Jojan Varghese (M.S. from Purdue)
Zachary Rasmussen (Undergraduate from Utah)
Rahul Golder (Undergraduate from IIT)
Soumick Sarker (Undergraduate from IIT)
Shraman Pal (Undergraduate from IIT)

Professional Activities

Reviewer for *Digital Chemical Engineering*	2025
Reviewer for *Applied Energy*	2025
Reviewer for *Computer & Chemical Engineering*	2025
Reviewer for *Journal of Aviation Technology and Engineering*	2025
Reviewer for *ACS In Focus*	2024
Reviewer for *Reviews in Chemical Engineering*	2024

Teaching Experience

Statistical Modeling and Quality Enhancement (*CHE-32000*), TA	Fall 2024
Momentum Transfer (*CHE-37700*), TA	Fall 2023

Skills

- Programming: Python, Julia, MATLAB
- Frameworks: PyTorch, SB3, Pyomo, OMLT, NumPy, Pandas, SciPy
- Solvers: Gurobi, CPLEX, Xpress
- Toolbox: Linux, vim, git